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Phase change materials in photovoltaic cells



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1 Preface

This report is written by a group of six Mechanical Engineering students from Delft University of Technology. The report is written for the second-year bachelor course 'Process Engineering and Thermodynamics', where knowledge from lectures is applied to specific technology projects. The project that this report discusses, is a study about the thermodynamical properties of Phase Change Materials.

To be able to fully understand the report, the reader must have a basic understanding of physical topics concerning thermodynamics, which can be found in chapter 3. Chapters 4 and 5 consist of a more thorough overview of our research and provide understanding of all our assumptions and calculations. Chapter 6 gives an overview of the results of the research. Readers only interested in the outcome of our research can best read the conclusion in chapter 7. Finally, chapter 8 contains a discussion.

We would like to thank our project supervisor Dr. Lorenzo Botto who provided us with the right counseling throughout the process. Additionally, we are thankful to Prof. dr. ir. J.T. Padding who provided us weekly with the theoretical knowledge to support our research.

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2 Summary

In context of the course Process Engineering and Thermodynamics all students of mechanical engineering at the faculty of 3Me had to do research about a thermodynamical process and improve it. In this report an attempt is made to enhance the efficiency of solar panels. This report describes our research and the final improvement.

First of all we started by analyzing the design of a solar panel and studying its main components. The main component of a solar panel is a photovoltaic cell, this conducts electricity at the end of the process. The PV cell consists of two semiconductors: a positive and a negative layer and a p-n junction layer in the middle. When the electrons in these layers receive the right amount of energy from incoming solar radiation and temperature, they jump to an excited state and starting moving between these layers. This is what induces the electricity that is generated. Unfortunately a significant amount of photons (from the solar radiation) needs to hit the solar panel to get the electrons into a certain band gap energy. If these electrons receive too little or too much energy a big part of the potentially harvested energy is converted to thermal energy and lost. This causes the temperature rise.

The efficiency depends on several parameters such as orientation of the solar cell, and research shows that temperature has a major influence on the efficiency of the solar cell, because the material properties change due to the temperature difference. When thermal energy is produced causes the efficiency of the solar panel to drop. A layer of Phase Change Material can be used as a temperature stabilizer. PCM functions as a heat sink since it can store or release energy when transferring aggregation states. If the right PCM is incorporated in the system it can significantly bump up the efficiency of the solar panel.

There are a couple critical variables that can optimize the process of generating electricity with a PV-PCM system. You can improve the volume of the PCM layer, the quality of the PCM layer, the thickness of the support material or the thickness of the PV cell. We have chosen to investigate in what climate a specific PV-PCM system would work the best. Each location has their own temperature and corresponding solar irradiance, which affect the efficiency of our system.

Next, to find the optimal temperature we gathered data about the temperature and solar irradiance of 10 locations with different climates all around the world. Subsequently a Python code was written containing formulas of every step taking place in the solar conducting process. By fixing all the variables except for the solar irradiance and temperature we were able to predict at what location and temperature our solar panel is most efficient.

In the end, when all the graphs were plotted it became clear that the highest energy efficiency is reached in relatively cold climate. In a location with low temperatures (eg. Reykjavik) the least amounts of energy are lost in the form of heat. Controversially, one often produced the biggest amounts of energy in locations with high solar irradiance (Nairobi) even if the efficiency on the spot is the lowest compared to colder locations with less solar irradiance.

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4 Introduction

In 2019 the world consumed 23,800 Terawatt-hours (TWh) of energy (Alves (2022)). In the same year, over 2% of the global electricity came from solar energy (Ritchie et al. (2022)), and this percentage is only expected to increase with time. This calls for new and advanced technologies to continue improving the efficiency of photovoltaic cells (PV cells) used in solar panels. The main problem faced when optimizing solar panels is that high solar radiation in combination with cool temperatures is preferred. Unfortunately high solar radiation frequently accompanies high temperatures, which then simultaneously results in a decreased efficiency.

This report provides us with an answer to the following question: How does the environmental temperature affect the efficiency of a PV cell? The research focuses on the optimization of PV cells, using phase change materials (PCM's). PCM functions as a cooling mechanism. They can store a lot of energy when transitioning between states of matter, and this process takes place at a constant temperature. Therefore their use in solar panels is beneficial for the efficiency of the PV cell: by absorbing and storing the excess in heat accompanying high solar radiation levels a lot of electricity can be generated with reduced energy loss. The main environmental parameters that affect this process are the air temperature and the solar irradiance. In order to be able to research the contribution of these variables to the efficiency of a solar panel we have created a programming code in Python that substantiates our findings.

This report holds the following structure: in Chapter 5, one can find a concise overview of the preliminary study that we carried out. Our preliminary study consists of general research and the summary of relevant articles. Chapter 6 holds the details of the situation we are going to research. Chapter 7 gives an overview of the Cost function we are analysing. A description of calculation and the used code is given in chapter 8 and the expected results and findings of our research are discussed in Chapter 9. Then in chapter 10 a prediction of the system's behaviour is given. The final results and analysis can be found in chapter 10 and at last the conclusion and recommendations are to be found in Chapter 11.

5 Preliminary study

5.1 The working mechanism of a photovoltaic cell

The photovoltaic cell of a solar panel consists of 3 layers: a positive layer, a negative layer and a p-n junction layer, which is the boundary between previously mentioned layers through which positive and negative photons pass. In the photovoltaic cell, you can find electrons bound at a low energy state. When these electrons receive extra energy, they enter a new state – known as the excited state – which allows them to break the bond and move. Electrons in the excited state can participate in conduction. The extra energy that elevates them to the excited state comes from two different sources; from light (sunlight) or from heat.

There is a certain band-gap energy. When the photons sent by the sun get less energy than the band-gap, they get converted to thermal energy. When they have too much energy they will be transferred into heat. Both cases aren't wanted. It's a challenge to get the photons into the exact band-gap energy and get them to convert to electrons in the PV-cell.

When light of the desired bandwidth (the band-gap) makes contact with the solar panel, the energy carried in the lightwaves' photons transfer to the electrons who then jump to a higher energy state. This enables the electrons to flow around between layers which results in an electric current. The conductor materials on the front and back of the solar panel then collect the electricity.

The optimal situation would be to put the solar panels in a relatively cool environment with lots of solar radiation. Unfortunately this optimal energy band-gap is quite small which results in a lower efficiency when the exact requirements aren't met. A possible solution to regulate temperature is to insert a PCM layer inside solar panels. As mentioned before, PCMs can store a lot of energy and can work as a cooling system in the solar panel. This results in more photons at the right energy level hence a higher efficiency. Other variables that optimize the process could be implementing an insulation layer or an anti-reflective layer on top of the panels to absorb more radiation.

5.2 Background information from relevant articles

What is latent heat?

Latent heat is the heat that is required to transform a unit mass of a material from the solid state to the liquid state. Thermal energy can be effectively stored by using latent heat storage systems with phase change materials (PCMs). It is beneficial to store energy because it reduces the discrepancy between the supply and demand of energy, as well as enhances the performance and reliability of energy systems.

There are three critical parts of a latent heat energy storage system: a proper material which changes material from solid to liquid at the anticipated melting point, a fitting surface for heat exchange to take place and a container that is compatible with the phase change material (Sharma et al. (2009)). An important quality of any PCM is that the heat absorbed and released occurs at an almost constant temperature.

Global analysis of photovoltaic energy output enhanced by phase change material cooling

A phase change material (PCM) is a layer in the solar panel with high latent heat capacity which acts as a heat sink and limits the peak temperature of the photovoltaic (PV) cell, absorbing heat that is transferred from a PV cell and thereby increasing efficiency since the solar cell efficiency is dependent on cell temperature.

PCMs are very effective as cooling mechanisms because other mechanisms such as air or water cooling are not beneficial in places where solar energy has the highest yield, where there is often a scarcity of water and/or high temperatures.

PCMs can either be isothermal during phase change or undergo a small phase change temperature range. Ideally, a PCM should have a small thermal expansion coefficient, high thermal conductivity, high latent heat of fusion and a high specific heat capacity. There are several ways to improve thermal conductivity,. Examples are attaching metal fins to the PCM to increase the contact surface area between the plate and the PCM, inserting high thermal conductivity particles into the PCM or micro-encapsulation (tiny particles or droplets are surrounded by a coating to give small capsules useful properties).

According to (Smith et al. (2014)) the PCM systems can control the temperature rise in a PV cell temperatures

during the daytime improving solar cell efficiency and electrical energy output. The best results are seen where an ideal PCM melting temperature for the location in question is used and the PCM melts fully over the course of the day and resolidifies in the evening. In this case the full latent heat content of the PCM is used and the PCM layer acts as thermal mass with a high effective heat capacity. It has been demonstrated that single junction silicon PV-PCM systems are not currently cost-effective, but it may be possible to achieve for technologies such as concentrating PV and multi-junction solar cells which reach higher temperatures and electrical outputs.

Performance enhancement of photovoltaic cells using phase change material (PCM) in winter

Increasing the operation temperature in photovoltaic cells (PV cell) worsens their performance. A possible thermal regulation system is a passive cooling technique using phase change material (PCM). Using a PCM regulation system allows the PV cell to operate at a constant temperature. Other possibilities are using an air based cooling system or pumping cooling water on the PV cell's surface. These options seem to be very effective but require an active energy input (in these specific examples a fan and a waterpump). This is of course a waste of energy.

PCMs have a high latent heat, which means the amount of potential energy storage is high. However the thermal conductivity is rather low; especially in their liquid phase. This deteriorates the cooling system when approaching higher temperatures. There has been a lot of research about incorporating an external material in the PCM to improve its thermal conductivity. However few investigations have been made on doing this with metal foams. The performance of PV-PCM systems under cold climate conditions has also not extensively been analyzed. This study reports the results of an experimental setup of a PCM with aluminium foam (AMF) during the winter in Benha city, Egypt. During the winter period the temperature distribution on the surface of the PV cell, open circuit voltage and output power was compared in a normal PV cell and in a PV cell with the PCM-AMF temperature regulation. The PCM-AMF PV cell brought the temperature down by a maximum of 15% and produced a maximum of 4.14% more power.

Improving the efficiency of photovoltaic cells using PCM infused graphite and aluminium fins

In a study done by Peter Atkin and Mohammed M. Farid (2015) the temperature changes of solar panels with the use of a graphite infused PCM and an external finned heat sink are examined (Atkin & Farid (2015)). They executed the experiment by comparing the panels in four cases: one without both potential optimizations, one with the graphite infused PCM, one with the external heat fins and the last one with both. Atkin and Farid came to the conclusion that the addition of the graphite infused PCM resulted in temporary temperature regulation and the external heat fins resulted in lower temperature peaks. In the end the solar panel with both the implemented PCM and the heat fins was the most efficient.

Optimization of a phase change material wallboard for building use

A computer simulation can be used for determining the optimal dimensions and use of a phase change material (PCM). The optimal thickness of a PCM can be found by a set of numerical experiments. The stored energy is optimal for a PCM thickness of around 10mm (see figure 2).

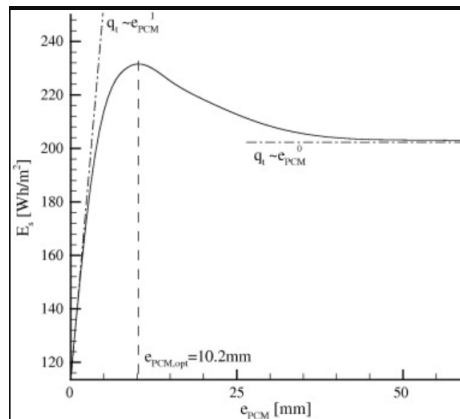


Figure 1: Results concerning the stored energy in a complete day E_s function of PCM thickness e_{PCM}

When also considering the following parameters: insulation thickness, outdoor air swing, phase difference and indoor air swing; a few compromises are made to obtain the optimal PCM thickness. This thickness is equal to

10mm (Kuznik et al. (2008)). The stockable energy can be compared with different insulating materials and can be found in figure 2.

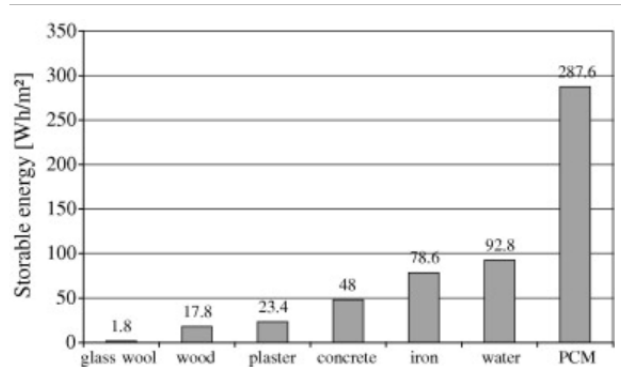


Figure 2: Maximum storable energy between 18 °C and 26 °C for 10mm of material during 24h

What are the variables that define the characteristics of a PCM?

PCMs are classified into three common types according to their structure Lawag & Ali (2022). These are organic, inorganic and eutectic PCMs. This is shown in an overview in the figure below (see figure 3)

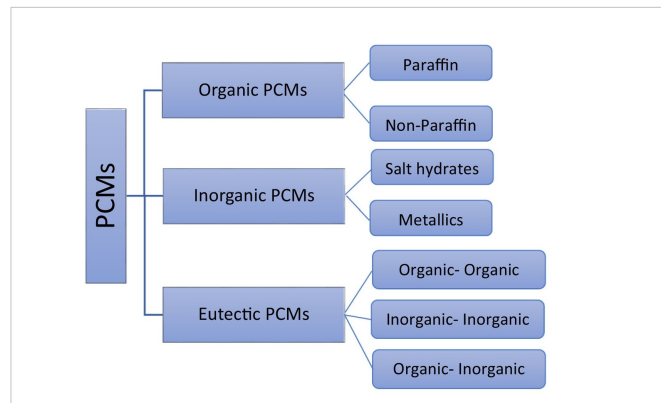


Figure 3: Overview of different types of PCM

Each PCM kind has their own advantages and disadvantages. When selecting a PCM for use in PV cells, it is important to consider the cost, sustainability, melting temperature and ratings of the material. Additionally, the type of material should be chosen based on the specific application and its specific requirements. There are multiple materials that can be used as PCMs. For example salt hydrates, organic compounds, metal hydrates or paraffin wax.

The cost of PCMs can vary depending on the type, material composition, and production method used. In general, organic PCMs are less expensive than inorganic PCMs, but they have a lower thermal stability.

In terms of sustainability, PCMs are generally considered to be a more sustainable option than other forms of energy storage, such as batteries. PCMs are non-toxic, non-flammable, and can be recycled.

The ratings of PCMs vary depending on the specific material used and the application. In general, PCMs with higher thermal stability tend to have higher ratings.

The most commonly used PCMs in photovoltaic cells are organic PCMs, such as Octadecane, erythritol, and acids which can be found in figure 4. These materials are relatively cheap, and have high thermal stability. They also tend to be more sustainable than other forms of energy storage.

Inorganic PCMs are also used in PV cells, and include materials such as salt hydrates, phase change salts,

and eutectic mixtures. These materials have higher thermal stability, but tend to be more expensive than organic PCMs. The thermophysical properties of some commonly used organic PCMs are given in the table below.

Organic PCM	h_f (kJ/kg)	T_m (°C)	K (W/m-K)	ρ (kg/m ³)	Ref
Paraffin wax	150–250	42–56	0.2	880–950	[26], [27]
Formic acid	277	7.8	0.30	1227	[26]
Adipic acid	275	152	–	1360	[28]
Erythritol	340	117	0.73	1450	[29], [30]
Palmitic acid	222	61	0.21	989	[22], [26]
Acetic acid	192	17	0.26	1214	[27], [33]
Acetamide	260	82	0.4	1160	[26]
Octadecane	241	28	0.26	774	[26]
Stearic acid	157	54	0.29	940	[27], [34]
Urea	250	134	0.8	1320	[30]
Lauric acid	212	44	0.22	66	[27], [33]

Figure 4: Organic PCMs and their properties

Octadecane seems like a favourable PCM, since Octadecane has a high latent heat, which is the amount of energy required to melt the material, an appropriate melting temperature and a relatively low cost (Lawag & Ali (2022)). Additionally, Octadecane is non-toxic, non-flammable and chemically stable.

6 Introduction to the model

A photovoltaic cell with a PCM layer is built up out of several different parts. Heat exchange takes place between each of these parts of the PV cell (illustrated in figure 5). The glass and anti-reflective coating are treated as one combined thermal mass due to their small heat capacity. An aluminum box sandwiches the PCM because it has shown that a highly conductive material is the most effective in a PV cell. Heat loss through the sides of the PCM can be neglected with respect to heat loss through the top and bottom, now covered by aluminum. (see figure 5).

To analyse this system an energy balance can be drawn up. The scheme consists of all the heat fluxes going in and out of our system (a few of these are depicted in figure 5)

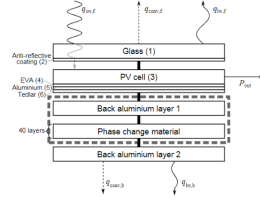


Figure 5: Energy balance system

To simplify the model and the energy balance system, a less complicated PV model was introduced. Some layers can be left out of consideration without having a significant impact on the results of the model. In the simplified model only 2 layers remain: the PV-cell and the PCM layer (see figure 6). Considering this different incoming energy forms can be distinguished; namely 'q convection', 'q shortwave' and 'q longwave'. The outgoing energy is considered to consist of two parts. A convective heat loss ('q convection') and the electrical power output. These are also shown in figure 6.

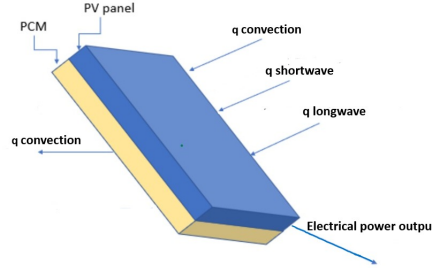


Figure 6: Simplified model of the PV cell with a schematic representation of the energy balance

We further simplify our input to the radiation of the sun (only considering an interval of the absorbable wavelengths). Between the PCM and the PV layer, energy is transferred by conduction. On the outgoing end we still have electricity (power out) as well as convection and unabsorbed solar radiation (heat losses). This simplified model is depicted in figure 7.

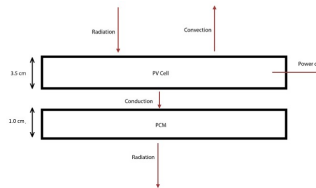


Figure 7: final simplified model of our solar panel with a schematic representation of the energy balance

7 Cost function

For this report we chose to focus on finding the optimal environment that results in the highest possible energy output for a PV-PCM installation. Ideally a PV cell would function best in an area where there is a lot of solar irradiance, but at moderate temperatures. However, in reality, a lot of solar irradiance is often accompanied by high temperatures. Therefore it is relevant to find out what the best balance between these parameters is.

7.1 Optimization Variable

In this research we have one optimization variable, which is the environmental temperature. We picked ten cities all over the world, and found their average temperature over the year 2022. These temperatures are the values that we will use to find an answer on our research question. Table 1) gives an overview of the cities and accessory temperatures and solar irradiances:

Table 1: Locations that are to be analysed

City	Temperature $^{\circ}C$	Solar irradiance W/m^2
Reykjavik	4,3	2,097
Oslo	5,9	2,749
Delft	10,9	3,0025
New York	11,9	3,948
Madrid	14,5	4,3575
Mexico city	16	5,392
Cape Town	16,4	5,369
Sydney	20	4,393
Nairobi	24	5,854
Singapore	26,7	4,501

As you can see, the data from Boxwell (2017) and Potter van Loon (2020) all varies in average temperature and solar irradiance. Using realistic data in our model makes our conclusion much more accurate.

7.2 Constraints

In order for our cost function to be complete, we must set a few constraints. These constraints help us simplify the model. First of all we set the material of the PCM to Octadecane Lawag & Ali (2022). Secondly, we model a day to be a period of eight hours starting from the moment the temperature rises. Additionally we defined the cross-sectional area of the PV-PCM combination to be $1 m^2$. We also set the thickness of the PCM at 10 millimeters. The solar panel's thickness is considered to be 35 millimeters. The manufacturing of the solar panels is left out of our calculation because it isn't relevant for our research.

7.3 Goal of cost function

The cost function is the amount of energy we can harvest from our PV-PCM system over a period of 8 hours. We want to maximize this energy output while minimizing our installation's price.

That is why one aspect of the cost function is the cost of the PCM. Although in our research model, we have chosen a specific PCM. This means all the variables that affect the PCM's price are fixed constraints. Therefore the cost is constant and it isn't relevant to further investigate this.

8 Description of calculations and code

8.1 Calculating P_{out}

Our cost function is based on how much energy our PV-PCM system can generate in an 8 hour day. This energy can be calculated with the following formula:

$$E = \int_{t=0h}^{t=8h} P_{out} dt$$

We want to maximize the energy generated and therefore we need to optimize the Power output (P_{out}) as well, which is the power generated by the PV cell. P_{out} can be calculated in 2 easy steps.

1) The efficiency can be calculated as followed:

$$\eta = \frac{P_{out}}{P_{in}}$$

2) This can be converted to:

$$P_{out} = \eta * P_{in}$$

Therefore the formula for E becomes:

$$E = \int_{t=0h}^{t=8h} \eta * P_{in} dt$$

P_{in} directly relates to the amount of solar irradiance the PV cell captures. We elaborate on the model we use to describe solar irradiance during the day is later in this paper (see 8.5 the python model).

The efficiency of the photovoltaic cell system depends on a lot of factors. The PV cell's temperature, the PCM's melting front, in- and outgoing heat fluxes are all considered in the calculations that are largely described in PCM notes.

As mentioned heat fluxes play an important role in this and there are several ways to look at them. The ones that have been considered in the model are discussed in the next subsection.

8.2 Relevant heat fluxes

First of all, for the heat flux, the convective heat flux to or from the air has to be considered.

$$q_{conv} = -h_{air}(T_{star} - T_{air})$$

We define the air temperature to be T_{air} , while T refers to the temperature of a point in the PCM. $h_{air}(W/[m^2K])$ is the heat transfer coefficient of air. The heat flux due to long wavelength radiation must also be considered:

$$q_{lw} = \sigma \cdot (\varepsilon_{air} * T_{sky}^4 - \varepsilon * T^4)$$

The emissivity of air is $\varepsilon_{air} = 0.95$; the emissivity of the material, ε must be checked on the PV cell itself. The Stefan-Boltzmann constant is $\sigma = 5.7 * 10^{-8} W/m^2/K^4$. Due to the contact between the PCM and the sky we have to calculate this temperature, using the unit Kelvin:

$$T_{sky} = 0.0552 * T_{air}^{1.5}$$

At last, the short wavelength flux is also relevant. It is expressed with:

$$q_{sw} = (1 - \eta) * \alpha * I$$

where $\alpha = 0.9$, is the absorbance of the PV cell and I is the irradiance (W/m^2) of short wavelength light. All together the heat flux at a point in our PV cell can be calculated according to this formula:

$$q_{tot} = q_{conv} + q_{lw} + q_{sw}$$

8.3 Modeling environmental parameters

As mentioned, our model depends on two environmental variables we are analysing: The air temperature and the solar irradiance. To do so, we have to describe said variables in a way that accurately represents them. We have chosen to do this using sinusoidal equations. These have been proven to be very reliable approximations, Mackintosh (2001).

The air temperature during the day is chosen to be modeled as a sine function with a variable amplitude and baseline.

$$T_{air} = B + A \cdot \sin\left(\frac{2 \cdot \pi}{24 \cdot 3600} \cdot t\right)$$

Here A will be our amplitude around the average temperature B. This will make the following formula.

$$T_{air} = T_{avg} + (T_{high} - T_{low})/2 \cdot \sin\left(\frac{2 \cdot \pi}{24 \cdot 3600} \cdot t\right)$$

Now we can input the average temperature to define a baseline and the high and low temperature to define the amplitude. This can be done for each of the 10 locations to model the air temperature of each location.

8.4 The Python model

The python code models our PV-PCM system in use at different locations. The model takes in to account the following factors: incoming irradiance due to the sun, irradiance due to longwave lengths, the irradiance due shortwave lengths and the out going irradiance from the PCM. These factors are modeled using the equation found in the PCM notes Tighe (n.d.). Note that the model follows a simplified version of the PV, PCM model. In reality the weather and propagation of the meltingfront follow very complex equations. In the code the following constants are used.

Table 2: Variables and constants used in model

Parameter	symbol	Value	Unit
Density	ρ	800	kg/m^3
Kelvin	-	273.15	Kelvin
Melting Temperature	T_{star}	25 + K	K
Start Temperature	T_{begin}	15 + K	K
PCM Thickness	D	0.01	m
Specific heat PCM	C_{pPV}	677	$J/(kg * K)$
Area	A	1	m^2
Stefan-Boltzmann coefficient	σ	$5.7 * 10^{-8}$	$W/(m^2 K)$
Emissivity of PV cell	ϵ_{pv}	$(0.9 + 0.95)/2$	-
Emissivity of sky	ϵ_{sky}	0.95	-
Emissivity of PCM	ϵ_p	0.8	-
Heat transfer coefficient of air	h_{air}	8.9	$W/m^2/K$
Thermal conductivity PV	k_{PV}	1.8	$w/(mK)$
Enthalpy	h_{PCM}	$230 * 10^3$	J
Thermal liquid conductivity PCM	k_l	0.2	$w/(mK)$
Reference temperature	T_{ref}	298	K
Efficiency at reference temperature	η	0.156	-
Beta constant	β	0.0045	1/ K

The values from each location are used in a for loop and the results are stored in a nested array. This results in a nested array with one parent array and 10 child arrays with each $8 \cdot 3600$ data points for every second during an 8 hour period for each of the 10 locations.

Next, 'for loops' are used to loop the data points through a set of equations found in the PCM notes Tighe (n.d.). This will result in the following nested arrays. Where each array again holds another 10 arrays with 8 hours of data points.

1. Temperature of environment

2. Temperature of PCM
3. Movement of the melting front
4. Energy produced
5. Efficiency

Using the datapoints from the models of the air temperature and the solar irradiance throughout the day the following ten step process is used to come to the final results:

1. Calculate irradiance of the sun.
2. Calculate the incoming total irradiance into the PV (Sun irradiance, long wavelength, short wavelength).
3. Calculate the temperature of the PV.
4. Calculate the outgoing irradiance from the PV to the PCM.
5. Calculate the outgoing irradiance from the PCM
6. Calculate the temperature of the PCM
7. Calculate the position of the melting front when the temperature of PCM reaches the melting temperature of the PCM.
8. Calculate the further increase of the PCM temperature after the PCM has fully melted.
9. Calculate the integral of the power out
10. Calculate the efficiency

All steps described above use a numerical integration method. It is important to note that the numerical method highly depends on the initial values given in the problem. When non-realistic values are given the numerical method will iterated until a value is reached that is realistic which can result in distortions with the graph.

9 Prediction of system behavior

9.1 Prediction of simulation based on temperature

Overall, the system should have a behaviour that results in the PCM increasing the efficiency of the photovoltaic cell by providing thermal energy storage.

Several studies on PCM materials in PV cells indicate that the generated energy in the form of electricity decreases when the temperature in the solar cell increases, as can be seen in figure 8 where the correlation between power output in the form of voltage (V) is shown against the electric current (A) at different temperatures.

In principle, it is preferable to have the ambient temperature at exactly the melting temperature, but because often high solar irradiation comes with high temperatures, concessions have to be made to find the optimum between low temperature and the highest possible radiation. In other words this paper aims to find a temperature as close as possible to the melting temperature together with as much radiation as possible. Because the aim for the temperature is to be as close as possible to the melting temperature, we actually want to have a constant temperature over a long period, and not sudden peaks in temperature. As these will melt the PCM very quickly in one run. After being completely melted the use of the PCM significantly decreases and it simply acts as an insulating layer in the solar panel. This means the longer it takes for the PCM to completely melt (within our 8h time frame), the more heat it can absorb, and thus the more efficiency is increased.

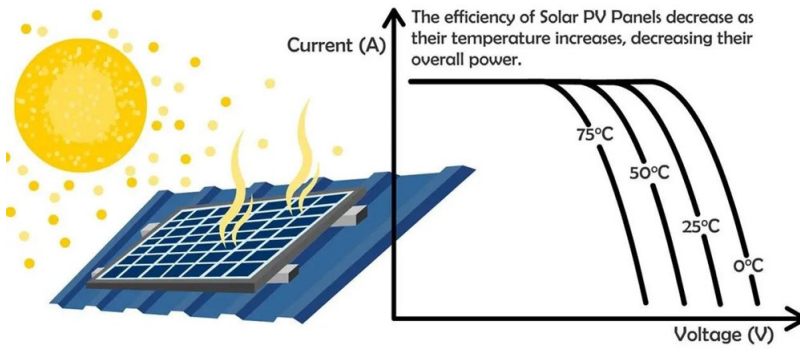


Figure 8: Correlation between power output and temperature in a PV cell

9.2 Comparison of the model to published literature

This paper aims to develop a python model that is used to predict the productivity of a photovoltaic system in combination with a PCM. To validate the different python model results, a comparison with published work is essential. The literature suggests that temperature has a strong influence on the efficiency of a photovoltaic (PV) cell. Higher temperatures can reduce the efficiency of a PV cell due to increased internal heating, which causes thermal losses, and due to decreased open-circuit voltage. Studies have shown that the efficiency of a PV cell decreases by 0.5-1 % for temperatures above 25°C and can reduce to half at higher temperatures. Therefore, keeping the PV cell at a lower temperature is beneficial for increasing its efficiency.

The efficiency can be improved thanks to the cooling mechanism of the use of PCM. Lower temperatures will ensure higher efficiency, and with that, relatively higher power output. Unfortunately lower temperatures often go hand in hand with a very little amount of solar irradiation. Based on this data and properties of a PV cell it can be predicted that even though the efficiency is reduced at higher temperatures, there is proportionally so much more solar irradiation present that the increased efficiency at low temperatures does not outweigh the high solar radiation at high temperatures.

10 Results and analysis

10.1 Results: an overview

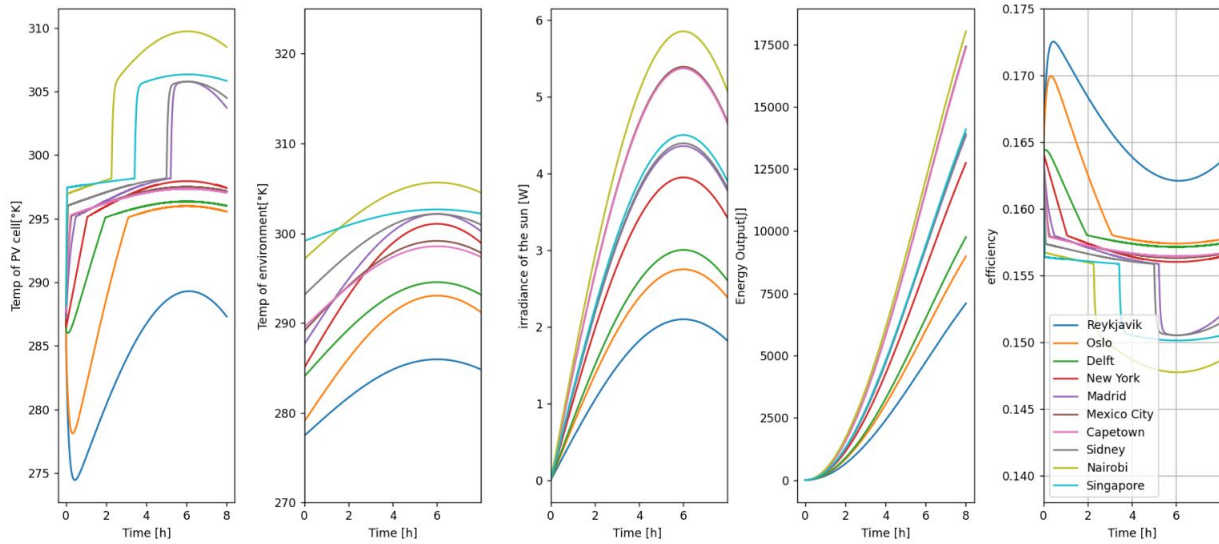


Figure 9: Overview of results

The figure above consists of the 5 graphs we are going to analyze and on which we will build our conclusion.

10.2 Analysis

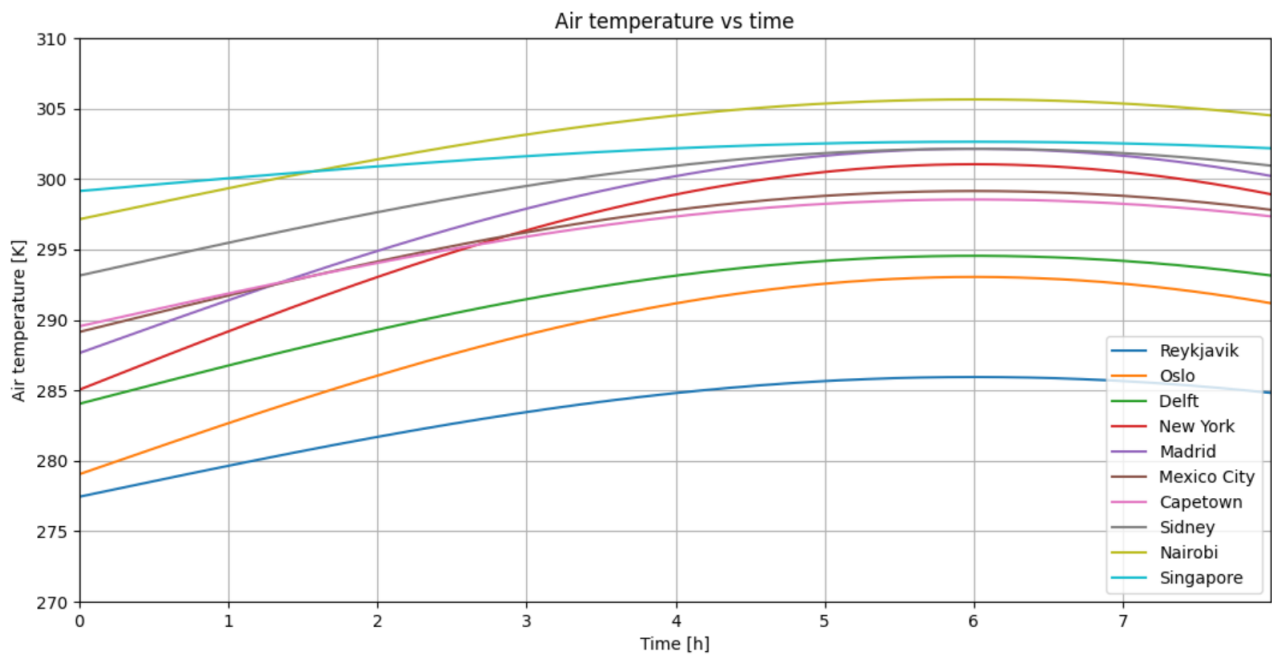


Figure 10: environmental temperature over time

In figure 10, over a period of 8 hours, the environmental temperature over time can be observed for each of the chosen cities. The temperatures that can be found in chapter 7 on page 10, are used as average temperature

and then modelled to a sinus function.

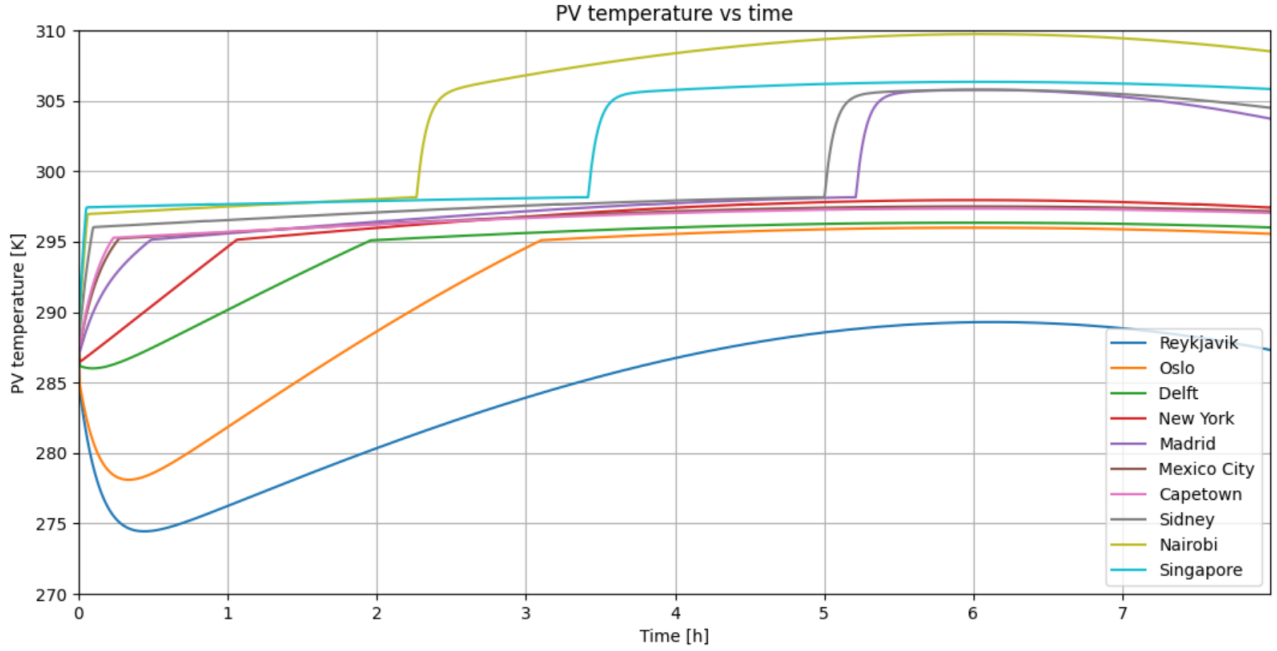


Figure 11: Temperature of PV cell over time

In figure 11, T_{PV} is modelled as a function of time. In chapter 8 one can find that the melting temperature of the PCM, T_{star} is 298.15 Kelvin. This explains why, at this specific instant, the temperature stocks for a period of time. The temperature will not change as long as the PCM material is melting. From this graph, it can therefore be found how long the PCM material will continue to melt; the longer this melting phase lasts, the lower the maximum temperature will turn out. This means when our PCM doesn't fully melt we get the most favorable situation, as this means our PV cell will never exceed it's optimal temperature. In doing so, we avoid the efficiency of the PV cell dropping drastically.

For example, the average temperature in Singapore is 26.7 °, which is the highest of all chosen cities. The line that represents Singapore stays horizontal relatively long at T_{star} , and therefore the efficiency drops only a couple of hours later than for example Nairobi.

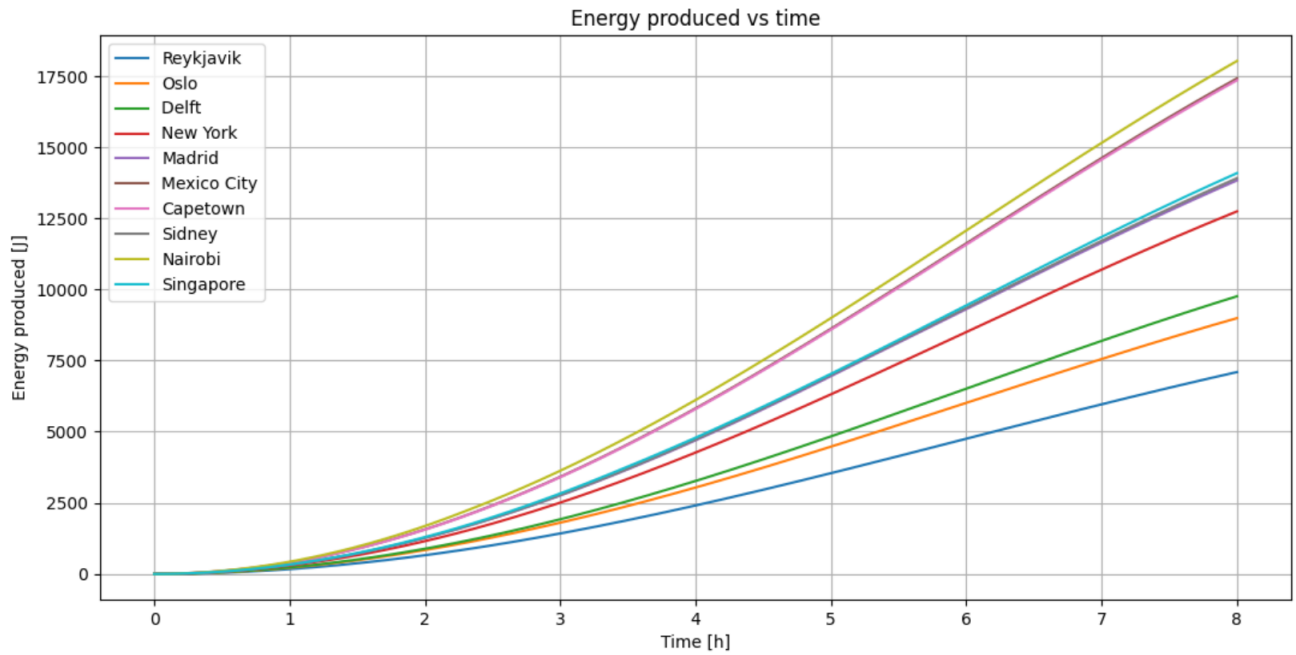


Figure 12: Energy produced over time

The third graph shows the relation of the produced energy at the different locations. These different locations have different temperatures and different corresponding solar irradiances which effects the P_{out} . Especially Nairobi and Capetown stand out in a positive way, these locations have produced the most energy. On the other hand Reykjavik has produced the least amount of energy by far. We will look further into this and compare this with their efficiency.

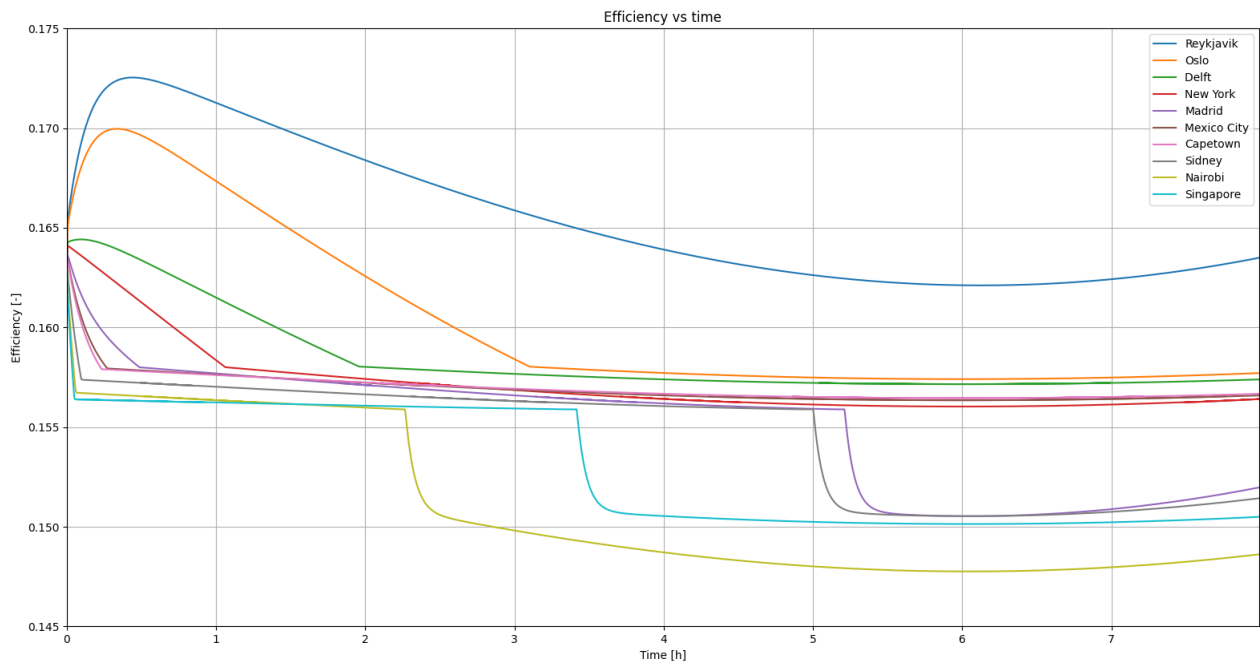


Figure 13: Efficiency over time

Finally, figure 13 shows us what the overall efficiency of the solar panels becomes in the end. As can be seen here, the total power efficiency of Reykjavik is the highest out of all our locations. This would seem illogical

as Reykjavik had the lowest Energy output as discussed before. But thanks to the environmental temperature globally staying the lowest in Reykjavik, the temperature of the PV also stays the lowest. This means that less energy gets lost to heat and the PV cell working more efficient.

Also the most significant efficiency drops happen when the PCM temperature exceeds it's melting temperature of 298.15 degrees, so it is preferred to keep the PCM under this boundary. When comparing this with the corresponding environmental temperature at the same time that boundary would be put at 302 degrees Kelvin.

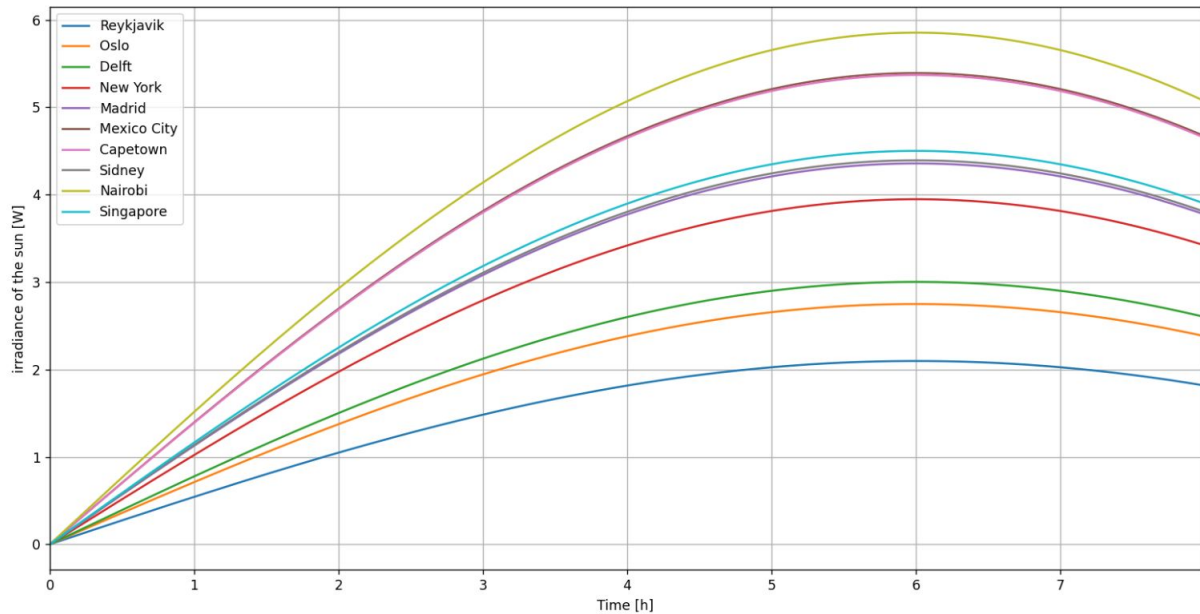


Figure 14: solar irradiance over time

Figure 13 tells us that efficiency of Nairobi is on average 0.015 lower than Reykjavik (with the highest efficiency). Controversially to this finding, figure 12 shows that Nairobi produces the most energy.

It becomes clear what happens when you take into account the solar irradiance hitting the solar panels over time (figure 14); the solar irradiance in Nairobi is almost 3 times as much as in Reykjavik. Consequently, even if the efficiency of Nairobi isn't nearly as high as in Reykjavik this is still getting compensated by the huge amounts of solar irradiance hitting on the solar panels.

In the end it is preferable to place your solar panel system in a climate with a low Environmental temperature but with huge amounts of solar irradiance to get the best efficiency. If this is too difficult to achieve the maximum environmental temperature drops should be put at 302 degrees Kelvin to keep the PCM under it's melting temperature and avoid a big efficiency drop.

Unfortunately these kinds of climates are difficult to find so if one is only concerned about producing as much energy as possible, one should look for locations with the most amounts of solar irradiance.

11 Conclusion

With our research we have been looking for the optimal conditions of a PV-PCM installation. We have chosen to focus on one parameter to optimize; the environmental temperature around a PV-PCM installation. And thus this report is aiming for an answer on the following question:

How does the environmental temperature affect the efficiency of a PV-PCM installation?

Two significant results can be seen from the graphs. First, locations with the highest temperature and the highest solar irradiance produce the most energy through a PV-PCM installation. But most importantly the locations with a lower environmental temperature show a higher PV cell efficiency. Therefore it can be concluded that when a PV-PCM installation needs to be optimized for a high efficiency it should be placed in locations that have a low average temperature.

12 Discussion

12.1 Discussing the results

For this report we researched the connection between the temperature in combination with solar irradiance, and the efficiency of a PV cell with a phase change material. We created a python model to substantiate our research, and we analysed our findings.

We were then able to draw a conclusion: the efficiency of a PV-PCM system is optimal in areas with low temperatures. However, the energy produced by the PV cell, is maximum in areas with large temperatures, where there is a lot of solar irradiance. This is somewhat what we expected, however, we expected the PCM to have a bigger influence on our results. The main goal of incorporating a phase change material in a PV cell is to improve the efficiency, especially for areas with a lot of solar irradiance, accompanied by high temperatures. Therefore we expected for the efficiency to be higher in the warmer cities, such as Singapore and Nairobi. It seems as though the PCM does not make a significant difference. The efficiency is calculated by dividing the P_{out} by the P_{in} , therefore in warm areas where there is a lot of sun, there is a high Power In. The Power Out, divided by a high Power In, leads to a smaller efficiency. On the other hand, if we look at the results of the P_{out} , it does match our expectations. We see that a lot of solar irradiance is accompanied with a high power output.

12.2 Possible errors and improvements

To find out whether the PCM has a lot of effect on the efficiency, we could create a second model with a regular PV cell, not making use of a PCM. This way it is easy to compare the efficiency of the cities and conclude if the PCM does indeed make a difference. In the future, it would be interesting to make a few alterations to make the model more realistic, and explore other options to make the solar cell more efficient.

First, we could look at multiple PCM thicknesses instead of 1 fixed thickness. We know that the thickness of the phase change material has an effect on the efficiency, and it would be interesting to see what this effect is, in order to find the optimal thickness.

Second, we could examine different PCM's. For this report we chose the material Octadecane, but as previously mentioned, there are multiple materials that can function as PCM, which might influence our result.

Furthermore, as explained in chapter 8, for the temperature we took the average temperature per day over a year of the specific city, and used this to model a sinus function over 8 hours. This may give more specific results than just using one temperature, but it is still a simplification compared to the actual temperature distribution over 8 hours.

Lastly, when looking at the P_{in} , we only took the solar irradiance in consideration, without distinguishing short and long wavelengths. Our model would be more accurate if we had specified a certain band gap such that the PV cell would only be able to absorb light within this range.

When looking specifically at the Python code, we feel like our model is relatively accurate and plausible. However, there are some aspects that we might have overseen, or assumptions that we have wrongfully made, which could have led to errors in the code. The latter could lead to different results which then effects our conclusion and therefore the outcome of our research.

References

- Alves, B. (2022). *Net electricity consumption worldwide in select years from 1980 to 2019*.
- Atkin, P., & Farid, M. M. (2015). Improving the efficiency of photovoltaic cells using pcm infused graphite and aluminium fins. *Solar Energy*, 114, 217-228. doi: <https://doi.org/10.1016/j.solener.2015.01.037>
- Boxwell, M. (2017). *The solar electricity handbook-2017 edition: A simple, practical guide to solar energy-designing and installing solar photovoltaic systems*. Greenstream Publishing.
- Kuznik, F., Virgone, J., & Noel, J. (2008). Optimization of a phase change material wallboard for building use. *Applied Thermal Engineering*, 28(11), 1291-1298. doi: <https://doi.org/10.1016/j.applthermaleng.2007.10.012>
- Lawag, R. A., & Ali, H. M. (2022). Phase change materials for thermal management and energy storage: A review. *Journal of Energy Storage*, 55, 105602. doi: <https://doi.org/10.1016/j.est.2022.105602>
- Mackintosh, L. (2001). Using trigonometric functions to model climate. *RSNZ Teacher Fellow, NIWA*.
- Potter van Loon, A. (2020). *World - solar irradiation and pv power potential map* (energydata.info, Ed.).
- Ritchie, H., Roser, M., & Rosado, P. (2022). Energy. *Our World in Data*. (<https://ourworldindata.org/energy>)
- Sharma, A., Tyagi, V., Chen, C., & Buddhi, D. (2009). Review on thermal energy storage with phase change materials and applications. *Renewable and Sustainable Energy Reviews*, 13(2), 318-345. doi: <https://doi.org/10.1016/j.rser.2007.10.005>
- Smith, C. J., Forster, P. M., & Crook, R. (2014). Global analysis of photovoltaic energy output enhanced by phase change material cooling. *Applied Energy*, 126, 21-28. doi: <https://doi.org/10.1016/j.apenergy.2014.03.083>
- Tighe, B. (n.d.). Notes on modeling phase change materials. *Tu delft*.

13 Appendix

```

1 import numpy as np
2 import matplotlib.pyplot as plt
3 '''OVERALL Constanten'''
4 K = 273.15
5 T_star = 25 + K          #[K] Smelttemp
6     #[deg C] Temperature air
7 Tbegin = 15 + K          #[deg C] Start temperature
8 D = 0.01                 #[m] thickness PCM
9 Cp_pv = 677              #Soortelijke warmte PV
10 A = 1
11 sigma = 5.7 * 10**-8      #Stefan-Boltzmann coefficient W/(m**2K**4)
12 e_pv = (0.9 + 0.95) / 2   #emissivity of air PV cell
13 e_sky = 0.95              #emissivity of the sky
14 e_p = 0.8                #emissivity of PCM
15 h_air = 8.9              #heat transfer coefficient of air
16 Cp_pv = (677 + 500) / 2   #?
17 k_pv = (1.8)              #Thermal conductivity pv
18 h_pcm_1 = 230E3          #[J] enthalpie
19 k_l_1 = 0.2               #[W/(m+FFFD)K] Thermal Conductivity liquid
20 Cp_l_1 = 2.0E3
21
22 '''define time space'''
23
24 t0 = 0.0
25 t1 = 8*3600
26 dt = 1                   #[s] Tijdsstap
27 t = np.linspace(t0,t1,round(1+((t1-t0)/dt)))
28
29 '''list of relevant weather parameters for different locations'''
30 Location = [['Reykjavik', 4.3,14,-3,2.097],
31             ['Oslo', 5.9,23,-5,2.749],
32             ['Delft ', 10.9,22,1,3.0025],
33             ['New York', 11.9,29,-3,3.948],
34             ['Madrid', 14.5,32,3,4.3575],
35             ['Mexico City', 16,25,5,5.392],
36             ['Capetown',16.4,29,11,5.369],
37             ['Sidney', 20,26,8,4.393],
38             ['Nairobi',24,28,11,5.854],
39             ['Singapore',26,32,25,4.501]]
40
41 ''' DIFFERENT AIR TEMPERATURE CURVES FOR DIFFERENT LOCATIONS'''
42 def Tairfunc(Tempavg, Temphigh,Templow,t):
43     A = (Temphigh - Templow)/2
44     T_air1 = (Tempavg + A * np.sin(((2*np.pi)/(3600 * 24))* t)) + K
45     return T_air1
46
47 '''DIFFERENT RADIATIONS OF THE SUN FOR DIFFERENT LOCATIONS'''
48 def qzonfunc(qavg, t):
49     qzon1 = qavg * np.sin(((2*np.pi)/(3600 * 24))* t)
50     return qzon1
51
52
53 '''relevant heat fluxes'''
54 def eta(Temp):
55     eta = 0.156 * (1 - 0.0045 * (Temp - 298))
56     return eta
57
58 def q_rad(T_air, T_pv):
59     T_sky = 0.0552 * T_air**1.5
60     q_rad = sigma * (e_sky * T_sky**4 - e_pv * T_pv**4)
61     return q_rad
62
63 def q_conv(T_air, T_pv):
64     q_con = h_air * (T_air - T_pv)
65     return q_con
66
67
68
69 def code(h_pcm, k_l, Cp_l):
70     '''making lists'''

```

```

71 x_star = np.ones((10,len(t))) * 0.0000001 #Melt Front
72 T0 = np.ones((10,len(t))) * Tbegin #Temperature beginning of PCM
73 TL = np.ones((10,len(t))) * Tbegin #Temperature end of PCM
74 q_in_pcm = np.ones((10,len(t))) #Flux in
75 T_pv = np.ones((10,len(t))) * (15 + 273.15)
76 q_in_pv = np.empty((10,len(t))) #total radiation [W/m**2]
77 q_zon1 = np.empty((10,len(t))) #radiation of the sun[W/m**2]
78 q_uit_pv = np.empty((10,len(t)))
79 Tair1 = np.empty((10,len(t)))
80 q_uit_pcm = np.empty((10,len(t)))
81 Energy_output = np.empty((10,len(t)))
82 Efficiency = np.empty((10,len(t)))
83 '''defining constants'''
84 Cp_s = Cp_l
85 k_s = k_l
86
87
88 '''main for loop'''
89 for i in range(len(t)-1):
90     '''parsing all the data from the location list'''
91     for j in range(len(Location)):
92         #air temp
93         Tair1[j][i] = Tairfunc(Location[j][1], Location[j][2],Location[j][3],i)
94         if q_zon1[j][i] <= 0:
95             q_zon1[j][i] = 0
96         q_zon1[j][i] = qzonfunc(Location[j][4], i)
97
98
99         q_in_pv[j][i] = q_rad(Tair1[j][i], T_pv[j][i]) + q_conv(Tair1[j][i], T_pv[j][i])
100         T_pv[j][i+1] = T_pv[j][i] + (q_in_pv[j][i] - q_uit_pv[j][i] + (1-eta(T_pv[j][i]))
* A * q_zon1[j][i])/ (Cp_pv)
101         q_uit_pv[j][i+1] = ((T_pv[j][i] - T0[j][i]) * k_pv) / 0.035
102         q_uit_pcm[j][i] = (sigma * (e_sky * (Tair1[j][i])**4 - e_p * TL[j][i]**4)) * -1
103
104
105
106     '''TEMPERATURE CURVE OF OUR PCM'''
107     #BEFORE THE MELTFRONT
108     if T0[j][i] < T_star:
109         q_in_pcm[j][i+1] = ((T_pv[j][i] - T0[j][i]) * k_s) / D
110         T0[j][i+1] = T0[j][i] + ((q_in_pcm[j][i] - q_uit_pcm[j][i]) / Cp_s)
111         TL[j][i] = T0[j][i]
112
113     #DURING THE MELTFRONT
114     else:
115         if x_star[j][i] < D or (T0[j][i] == T_star and q_zon1[j][i] <= 0):
116             T0[j][i] = T_star
117             q_in_pcm[j][i] = ((T_pv[j][i] - T0[j][i]) * k_l) / D
118             x_star[j][i+1] = x_star[j][i] + ((q_in_pcm[j][i] - q_uit_pcm[j][i]) /
h_pcm) * dt
119             T0[j][i+1] = ((q_in_pcm[j][i] * x_star[j][i+1]) / k_l) + T_star
120             TL[j][i+1] = -((q_uit_pcm[j][i] * (D - x_star[j][i+1])) / k_s) + T_star
121
122     #AFTER THE MELTFRONT
123     else:
124         x_star[j][i+1] = D
125         q_in_pcm[j][i+1] = ((T_pv[j][i] - T0[j][i]) * k_l) / D
126         T0[j][i+1] = T0[j][i] + (q_in_pcm[j][i] - q_uit_pcm[j][i]) / Cp_s
127         TL[j][i+1] = T0[j][i] + (q_in_pcm[j][i] - q_uit_pcm[j][i]) / Cp_s
128
129
130     '''Integral to calculate the energy production in the PV cell'''
131
132     Energy_output[j][i+1] = Energy_output[j][i] + ((eta(T_pv[j][i]) * q_zon1[j][i]) * A)
* dt
133     Efficiency[j][i] = eta(T_pv[j][i])
134
135     return Energy_output, T0, TL, T_pv, Efficiency, x_star, Tair1, q_zon1
136
137 Energy_output, T0, TL, T_pv, Efficiency, x_star, Tair1, q_zon1 = code(h_pcm_1, k_l_1, Cp_l_1)
138
139
140 fig, ax = plt.subplots(1, 5, figsize=(8, 3))

```

```

141 for j in range(len(Location)):
142     ax[0].plot(t/3600, T_pv[j], label=Location[j][0])
143     ax[1].plot(t/3600, Tair1[j], label=Location[j][0])
144     ax[2].plot(t/3600, q_zon1[j], label = Location[j][0])
145     ax[3].plot(t/3600, Energy_output[j], label=Location[j][0])
146     ax[4].plot(t/3600, Efficiency[j], label=Location[j][0])
147 ax[0].set_xlabel('Time [h]')
148 ax[1].set_xlabel('Time [h]')
149 ax[1].set_xlim([0, 7.999])
150 ax[1].set_ylim([270, 325])
151 ax[2].set_xlabel('Time [h] ')
152 ax[2].set_xlim([0, 7.999])
153 ax[3].set_xlabel('Time [h]')
154 ax[4].set_xlabel('Time [h]')
155 ax[4].set_xlim([0, 7.999])
156 ax[4].set_ylim([0.138, 0.175])
157 ax[0].set_ylabel('Temp of PV cel[°C]')
158 ax[1].set_ylabel('Temp of environmen[°C]')
159 ax[2].set_ylabel('irradiance of the sun [W]')
160 ax[3].set_ylabel('Energy Output[J]')
161 ax[4].set_ylabel('efficiency')
162 plt.legend()
163 plt.grid()
164
165 # plt.show()
166
167 # #single plotting
168 # for j in range(len(Location)):
169 #     plt.plot(t/3600, q_zon1[j], label = Location[j][0])
170 # plt.xlabel('Time [h] ')
171 # plt.xlim([0, 7.999])
172 # plt.ylabel('irradiance of the sun [W]')
173 # plt.legend()
174 # plt.grid()
175 # plt.show()

```